

Predicting the Success of Bank Telemarketing for Selling Long-term Deposits

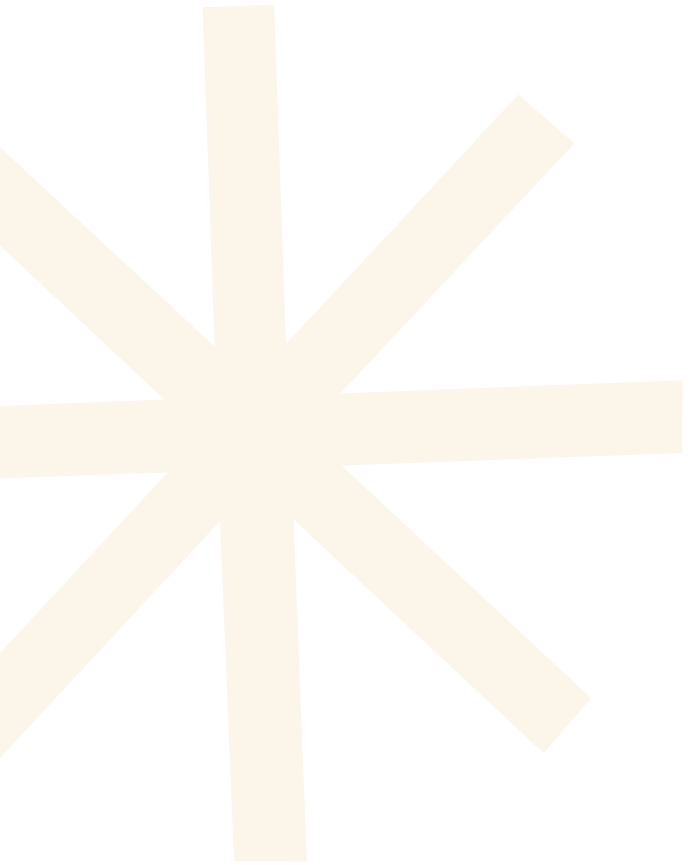
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GitHub link:

<https://github.com/Anna-Ikeda/Predicting-the-Success-of-Bank-Telemarketing-for-Selling-Long-term-Deposits.git>





Agenda Overview

- 01 Introduction of our research paper
- 02 The methodology
- 03 Contributions 1:
Upgraded version of the existing models
- 04 Contributions 2:
Adding different models
- 05 Contributions 3:
Testing the models of the paper on Shopper Purchasing Intention Dataset
- 06 Challenges and learning

Introduction of our research paper



Predicting the Success of Bank Telemarketing for Selling Long-term Deposits: An Application of Machine Learning Algorithms(2021)

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https://www.researchgate.net/publication/352755139_Predicting_the_Success_of_Bank_Telemarketing_for_Selling_Long-term_Deposits_An_Application_of_Machine_Learning_Algorithms

Introduction of the problem being solved

Whether telemarketing is effective
for encouraging customers
to subscribe to long-term bank deposits

- Few studies have predicted telemarketing success for long-term deposits using ML
- This research provided **insightful information to banks** for improving their telemarketing strategies for long-term bank deposits



Introduction of the dataset

Overview

- Our dataset is related with **direct marketing campaigns (phone calls) of a Portuguese banking institution**

- It collected from the UC Irvine Machine Learning Repository

<https://archive.ics.uci.edu/dataset/222/bank+marketing>

Number of Variables

- 21 features

Number of Rows

- 4,119 instances

Target and descriptive features

- Target feature: "y (Label)"
- Descriptive features: The rest of the features

Table 1. Features of the dataset

S. No	Features	Description of Features
1	Age	Age of the client
2	Job	Type of client's job
3	Marital	Client's marital status
4	Education	Highest education of the client
5	Default	Client credit
6	Housing	Housing loan
7	Loan	Personal loan
8	Contact	Type of contact communication with the client
9	Month	Last month of the year contracting to the client
10	Day of week	Last day of the week contracting to the client
11	Duration	Duration of client contact
12	Campaign	Number of contacts performed during the campaign and for this client
13	Plays	Number of days elapsed after the client's last visit
14	Previous	Number of contacts performed before this campaign and for this client
15	Poutcome	Outcome of the previous marketing campaign
16	Emp.var.rate	Employment variation rate
17	Cos.price.IDX	Consumer price index
18	Cons.conf.IDX	Consumer confidence index
19	Euribor3m	Euribor 3-month rate
20	Nr. employed	Number of employees
21	Label	Client subscription

Source: <https://archive.ics.uci.edu/ml/datasets/Bank+Marketing>

Methodology

(Reproduced steps)

01

Data Acquisition

UCI Machin Learning Repository.
21 attributes
4119 records

02

Exploratory data analysis

Check data's shape
Unique labels of the target variable.
Features data types
Data's descriptive statistics

03

Feature engineering

Convert categorical variable into
numeric
Apply the standard scaler technique

Methodology

(Reproduced steps)

04

Build model

Split the data into 90% train data and 10% test data. Also it was applied Standar scaling and 10-fold cross validation

- Logistic Regression
- Decision Tree
- Random Forest
- Gaussian Naive Bayes
- Support Vector Machine



Split dataset (X,Y)

X-train

Y-train

X-test

Y-test

Build the model

Y-pred

05

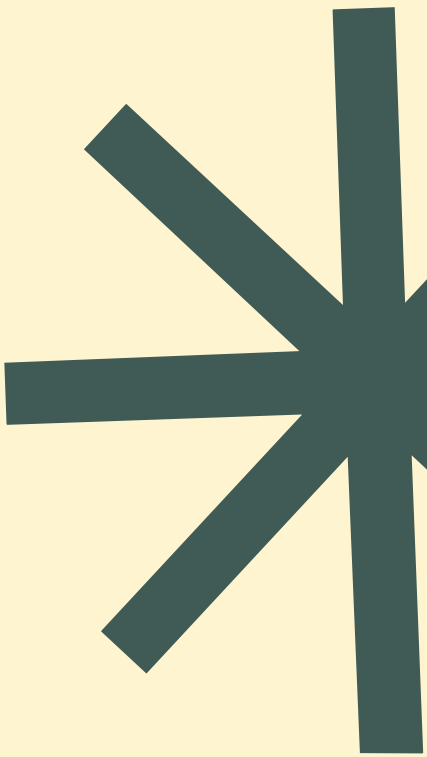
Evaluation metrics

Evaluate the performance of the models through confusion matrix

- Accuracy
- Precision
- Recall
- F1 Score
- AUROC Score

Contributions

Upgraded
version



Classification Report of Logistic Regression

The Result from our reproduced code

```
Classification Report:
              precision    recall  f1-score   support

   No         0.93         0.98         0.96         367
   Yes         0.75         0.40         0.52          45

 accuracy          0.92         412
 macro avg         0.84         0.69         0.74         412
 weighted avg      0.91         0.92         0.91         412
```

After apply hyperparameter tuning

```
Logistic Regression Classification Report:
              precision    recall  f1-score   support

    0         0.93         0.98         0.95         367
    1         0.73         0.36         0.48          45

 accuracy          0.92         412
 macro avg         0.83         0.67         0.72         412
 weighted avg      0.90         0.92         0.90         412
```

After apply SMOTE

```
Logistic Regression Classification Report:
              precision    recall  f1-score   support

    0         0.98         0.87         0.92         367
    1         0.46         0.89         0.61          45

 accuracy          0.87         412
 macro avg         0.72         0.88         0.77         412
 weighted avg      0.93         0.87         0.89         412
```

The Result from the research paper

```
Classification Report of Logistic Regression:
              precision    recall  f1-score   support

    0         0.9471         0.9728         0.9598         368
    1         0.7059         0.5455         0.6154          44

 accuracy          0.9272         412
 macro avg         0.8265         0.7591         0.7876         412
 weighted avg      0.9213         0.9272         0.9230         412
```


Decision Tree - Classification report

The results from our reproduced code

The results from the research paper

Classification Report:

	precision	recall	f1-score	support
No	0.93	0.97	0.95	367
Yes	0.59	0.38	0.46	45
accuracy			0.90	412
macro avg	0.76	0.67	0.70	412
weighted avg	0.89	0.90	0.89	412

After apply hyperparameter tuning

Decision Tree Classification Report:

	precision	recall	f1-score	support
0	0.94	0.95	0.94	367
1	0.52	0.49	0.51	45
accuracy			0.90	412
macro avg	0.73	0.72	0.72	412
weighted avg	0.89	0.90	0.89	412

After apply SMOTE

Decision Tree Classification Report:

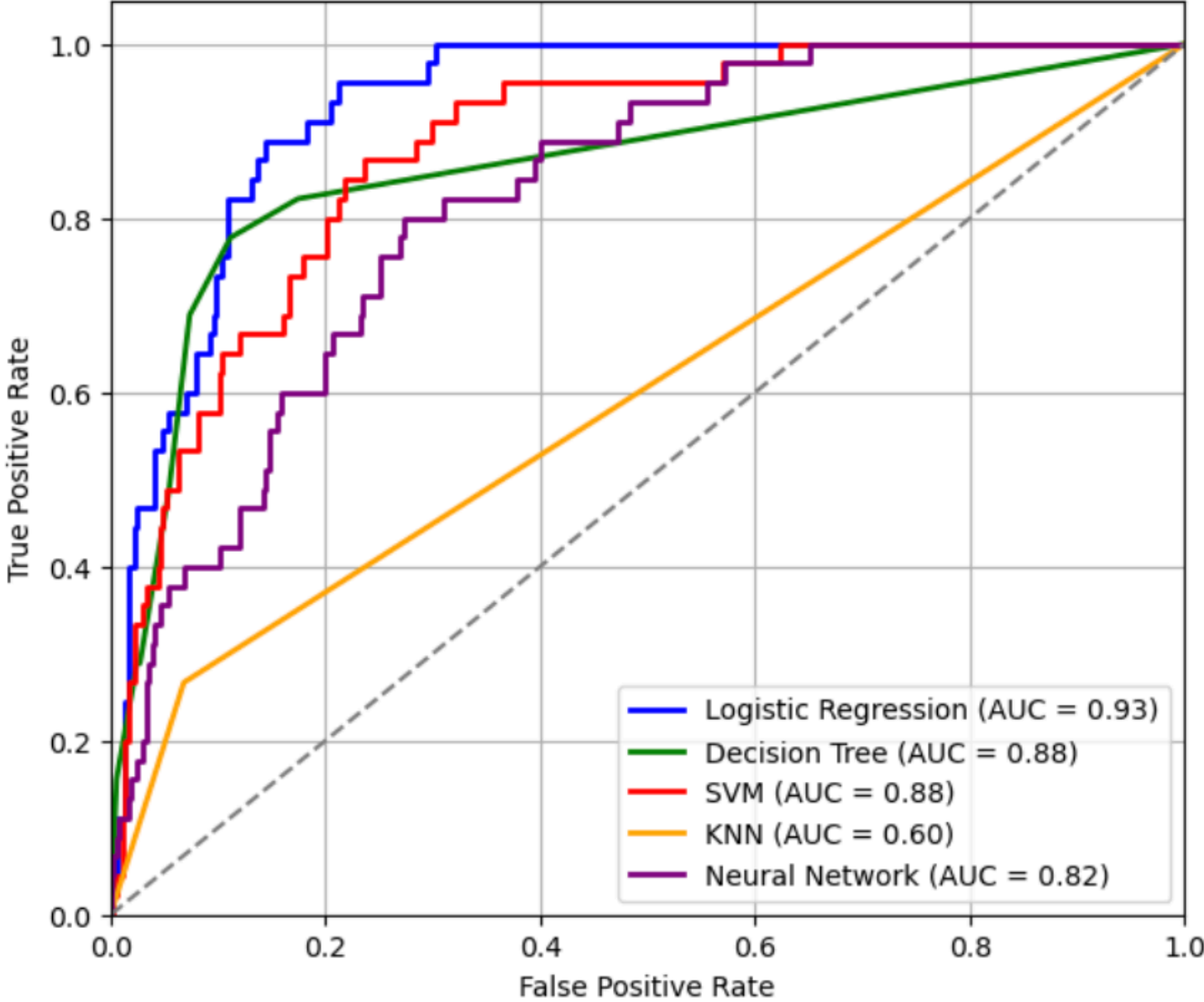
	precision	recall	f1-score	support
0	0.97	0.87	0.92	367
1	0.43	0.78	0.55	45
accuracy			0.86	412
macro avg	0.70	0.82	0.73	412
weighted avg	0.91	0.86	0.88	412

Classification Report of DT:

	precision	recall	f1-score	support
0	0.9387	0.9565	0.9475	368
1	0.5676	0.4773	0.5185	44
accuracy			0.9053	412
macro avg	0.7531	0.7169	0.7330	412
weighted avg	0.8990	0.9053	0.9017	412

Adding different models

AUC-ROC Curve Comparison: Logistic Regression vs Decision Tree vs SVM vs KNN vs Neural Network



The new models (SVM, KNN, and Neural Networks) did not outperform the original models (Logistic Regression and Decision Tree)

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	91.99%	75.00%	40.00%	52.17%	92.98%
Decision Tree	89.56%	52.38%	48.89%	50.57%	85.76%
SVM	89.56%	53.12%	37.78%	44.16%	87.93%
KNN	85.92%	32.43%	26.67%	29.27%	59.93%
Neural Networks	88.83%	48.39%	33.33%	39.47%	82.10%

The results shows us **low Recall** across all classifiers → Using **SMOTE**



Adding different models

SVM

```
SVC Evaluation
-----
Accuracy      : 89.56%
Precision     : 53.12%
Recall        : 37.78%
F1 Score      : 44.16%
AUC Score     : 87.93%
Confusion Matrix:
[[352  15]
 [ 28  17]]

Classification Report:
              precision    recall  f1-score   support

      No         0.93      0.96      0.94       367
      Yes         0.53      0.38      0.44        45

   accuracy              0.90       412
  macro avg         0.73      0.67      0.69       412
 weighted avg         0.88      0.90      0.89       412
```

SVM with SMOTE

```
SVC with SMOTE Evaluation
-----
Accuracy      : 84.95%
Precision     : 40.86%
Recall        : 84.44%
F1 Score      : 55.07%
AUC Score     : 92.98%
Confusion Matrix:
[[312  55]
 [  7  38]]

Classification Report:
              precision    recall  f1-score   support

      No         0.98      0.85      0.91       367
      Yes         0.41      0.84      0.55        45

   accuracy              0.85       412
  macro avg         0.69      0.85      0.73       412
 weighted avg         0.92      0.85      0.87       412
```

	Accuracy	Precision	Recall	F1-score	ROC-AUC
SVM	89.56%	53.12%	37.78%	44.16%	87.93%
SVM with SMOTE	84.95%	40.86%	84.44%	55.07%	92.98%

Adding different models

KNN

```
KNN Evaluation
-----
Accuracy      : 85.92%
Precision      : 32.43%
Recall         : 26.67%
F1 Score       : 29.27%
AUC Score      : 59.93%
Confusion Matrix:
[[342  25]
 [ 33  12]]

Classification Report:
              precision    recall  f1-score   support

      No         0.91         0.93         0.92         367
      Yes         0.32         0.27         0.29          45

   accuracy                0.86         412
  macro avg         0.62         0.60         0.61         412
 weighted avg         0.85         0.86         0.85         412
```

KNN with SMOTE

```
KNN with SMOTE Evaluation
-----
Accuracy      : 79.61%
Precision      : 33.04%
Recall         : 84.44%
F1 Score       : 47.50%
AUC Score      : 82.47%
Confusion Matrix:
[[290  77]
 [  7  38]]

Classification Report:
              precision    recall  f1-score   support

      No         0.98         0.79         0.87         367
      Yes         0.33         0.84         0.47          45

   accuracy                0.80         412
  macro avg         0.65         0.82         0.67         412
 weighted avg         0.91         0.80         0.83         412
```

	Accuracy	Precision	Recall	F1-score	ROC-AUC
KNN	85.92%	32.43%	26.67%	29.27%	59.93%
KNN with SMOTE	79.61%	33.04%	84.44%	47.50%	82.47%

Adding different models

Neural Networks

Neural Network Evaluation

Accuracy : 88.83%
Precision : 48.39%
Recall : 33.33%
F1 Score : 39.47%
AUC Score : 82.10%

Confusion Matrix:

```
[[351 16]
 [ 30 15]]
```

Classification Report:

	precision	recall	f1-score	support
No	0.92	0.96	0.94	367
Yes	0.48	0.33	0.39	45
accuracy			0.89	412
macro avg	0.70	0.64	0.67	412
weighted avg	0.87	0.89	0.88	412

Neural Networks with SMOTE

Neural Network with SMOTE Evaluation

Accuracy : 88.35%
Precision : 47.17%
Recall : 55.56%
F1 Score : 51.02%
AUC Score : 90.48%

Confusion Matrix:

```
[[339 28]
 [ 20 25]]
```

Classification Report:

	precision	recall	f1-score	support
No	0.94	0.92	0.93	367
Yes	0.47	0.56	0.51	45
accuracy			0.88	412
macro avg	0.71	0.74	0.72	412
weighted avg	0.89	0.88	0.89	412

	Accuracy	Precision	Recall	F1-score	ROC-AUC
Neural Networks	88.83%	48.39%	33.33%	39.47%	82.10%
Neural Networks with SMOTE	88.35%	47.17%	55.56%	51.02%	90.48%

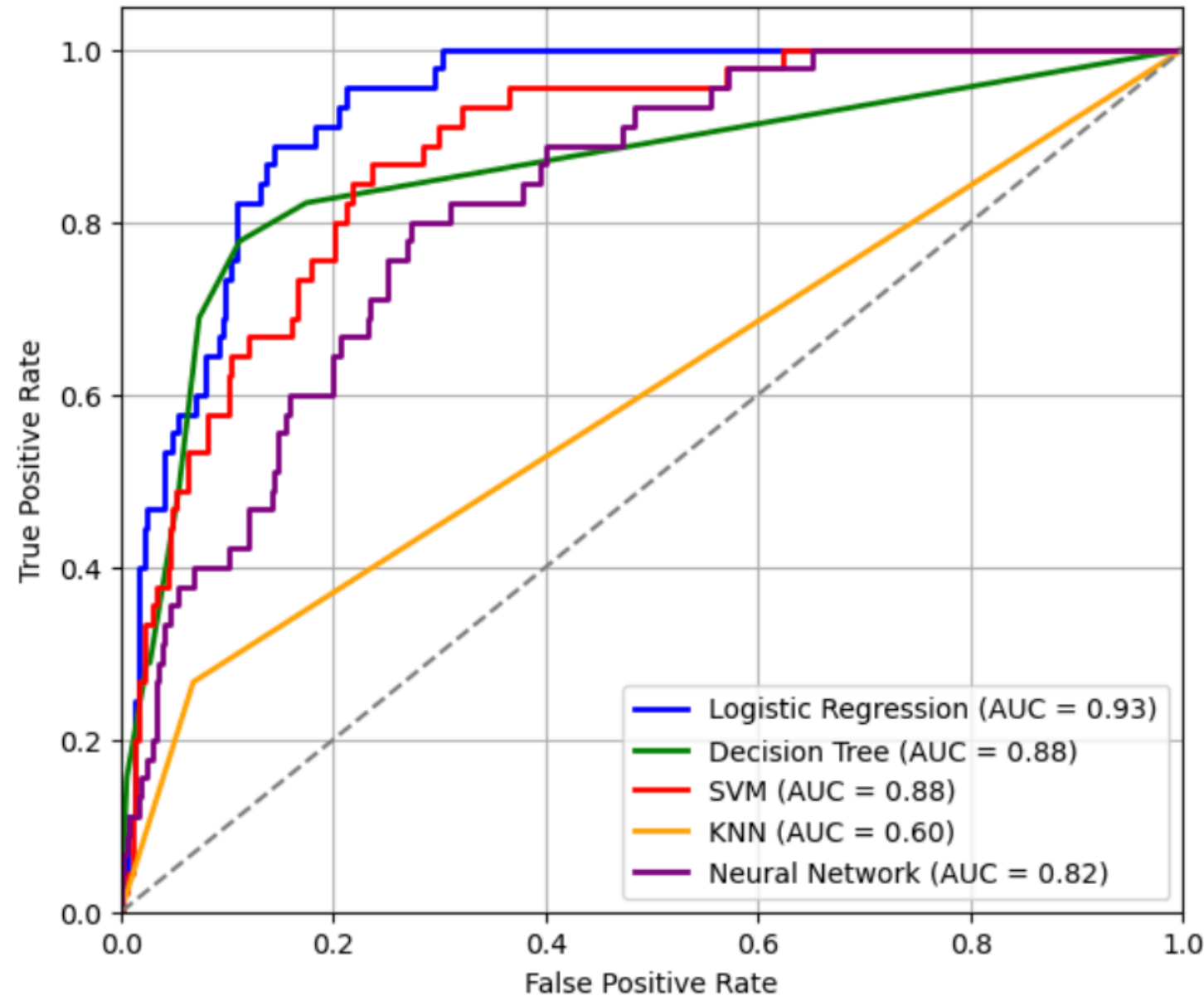
Model	SMOTE	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	No	91.99%	75.00%	40.00%	52.17%	92.98%
	Yes	87.14%	45.45%	88.89%	60.15%	94.28%
Decision Tree	No	89.56%	52.38%	48.89%	50.57%	85.76%
	Yes	87.62%	46.25%	82.22%	59.20%	89.33%
SVM	No	89.56%	53.12%	37.78%	44.16%	87.93%
	Yes	84.95%	40.86%	84.44%	55.07%	92.98%
KNN	No	85.92%	32.43%	26.67%	29.27%	59.93%
	Yes	79.61%	33.04%	84.44%	47.50%	82.47%
Neural Networks	No	88.83%	48.39%	33.33%	39.47%	82.10%
	Yes	88.35%	47.17%	55.56%	51.02%	90.48%

The best model: **Logistic Regression with SMOTE**

It achieved the highest recall (88.89%) and the highest ROC-AUC (94.28%), showing that it is very effective at identifying potential customers

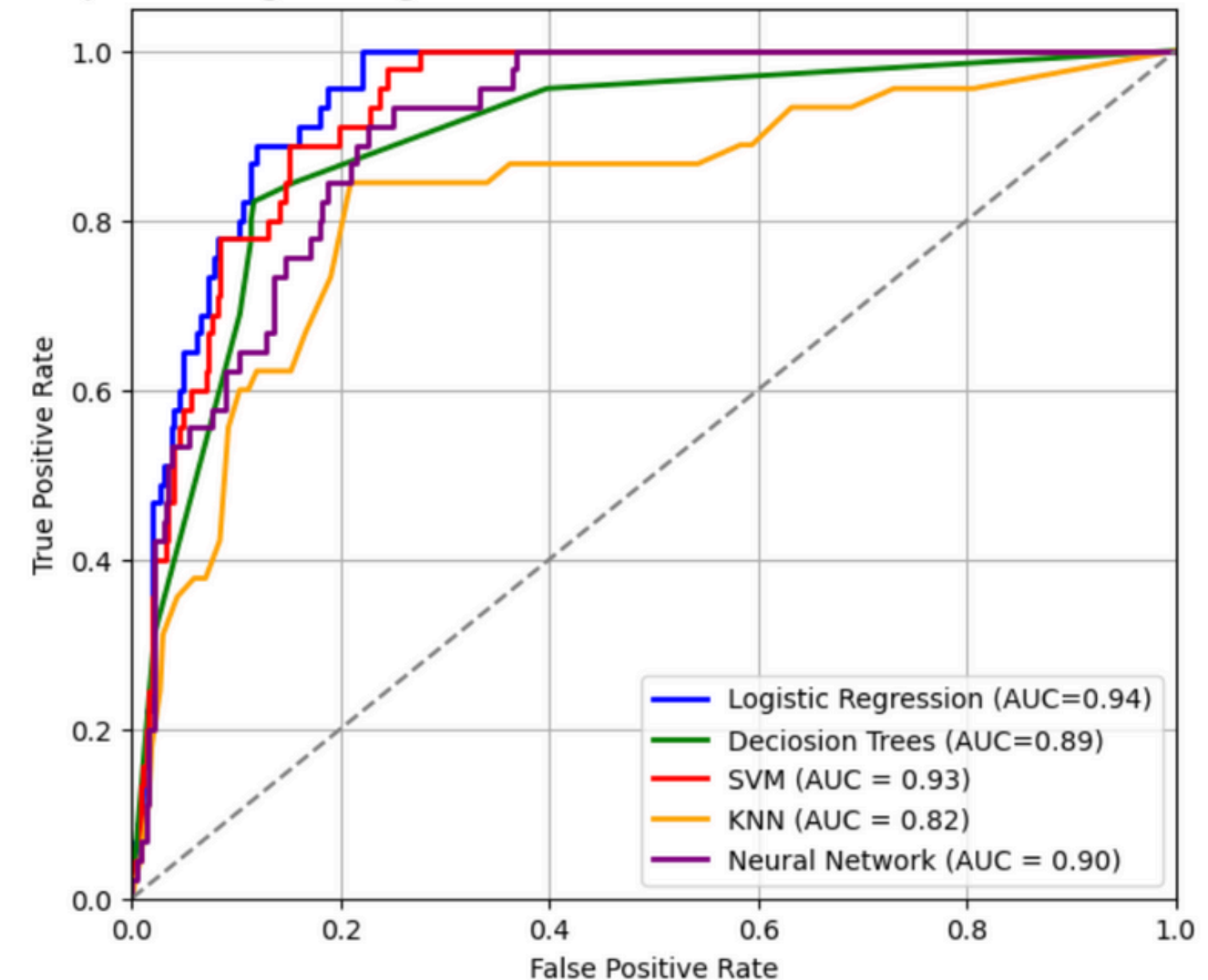
Whether our contributions have improved the model performance or not

AUC-ROC Curve Comparison: Logistic Regression vs Decision Tree vs SVM vs KNN vs Neural Network



As a result, **our work strengthened the original study** by confirming that its conclusions remain valid after testing additional models

AUC-ROC Curve Comparison: Logistic Regression vs Decision Tree vs SVM vs KNN vs Neural Network with SMOTE



Analysis of Shopper Purchasing Intention Dataset

Baseline Evaluation Metrics

Model	Accuracy	Precision	Recall	F1-Score	AUC
Logistic Regression	88.08%	72.92%	36.65%	48.78%	90.18%
Decision Tree	89.94%	73.76%	54.45%	62.65%	91.73%
SVM (Linear)	87.83%	69.90%	37.70%	48.98%	87.59%

Note: Baseline models show high accuracy but suffer from very low recall on the purchaser class.

SMOTE Evaluation Metrics

Model (with SMOTE)	Accuracy	Precision	Recall	F1-Score	AUC
Logistic Regression	85.16%	51.41%	76.44%	61.47%	90.92%
Decision Tree	86.29%	53.74%	82.72%	65.15%	91.37%
SVM (Linear)	87.02%	56.18%	73.82%	63.80%	91.74%

SMOTE significantly improves recall and F1-score across all models by balancing the target distribution.

ROC Curves: Baseline Models

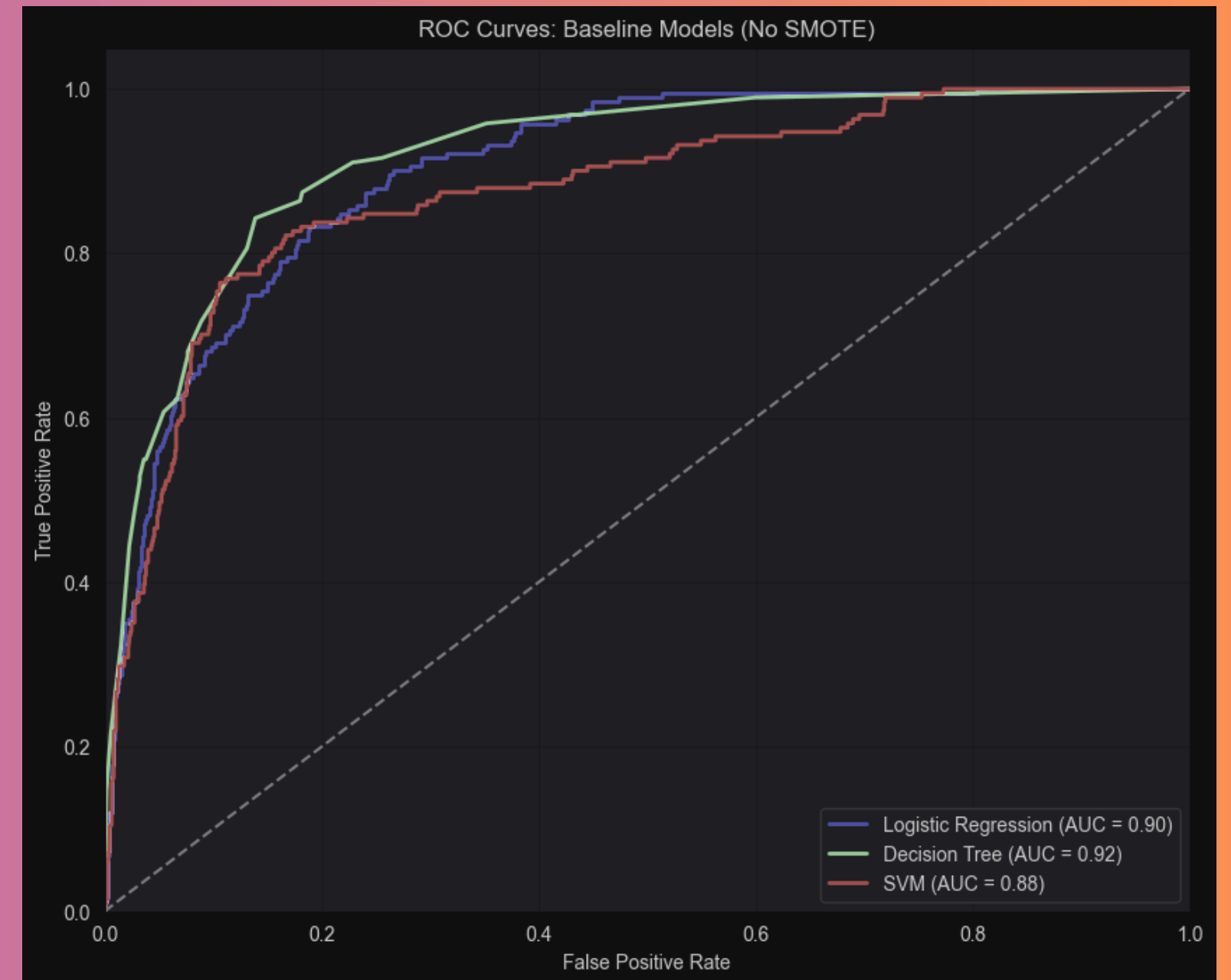
Initial Diagnostics

The baseline models demonstrate strong overall discriminatory power with AUC values between 0.87 and 0.92.

Decision Tree: Leads with an AUC of 0.92.

SVM: Slightly lags at 0.88.

The steep curve suggests models identify "non-purchasers" very easily, but struggle with "purchasers."



ROC Curves: Models with SMOTE

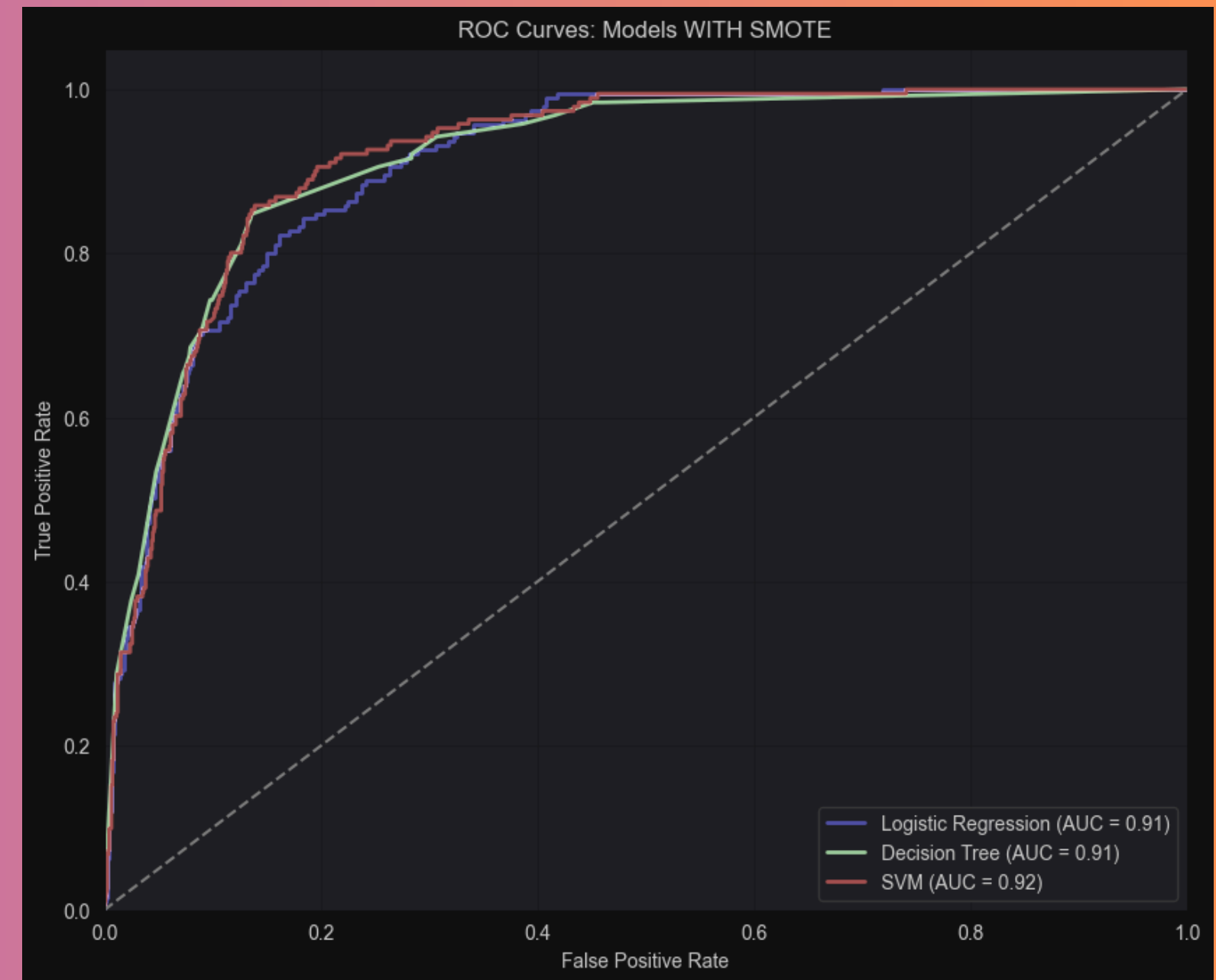
Oversampling Impact

Oversampling the minority class leads to more stable curves and enhanced performance for sensitive models.

SVM Gains: AUC increased significantly to 0.92.

Balance: Curves are more closely aligned, indicating consistent performance.

True Positive Rate (Recall) is optimized at the cost of a higher False Positive Rate.



Challenges

01

Class Imbalances

02

**Computational
Complexity of SVM**

03

Lack of domain expertise

04

Team Collaboration



Future Scope

01

Alternative Imbalance Handling Methods

- Other resampling techniques such as ADASYN

02

Feature Selection

- Applying feature selection or dimensionality reduction techniques could improve model performance and reduce computational cost



Thank You

